

Networking

Definition

Networking is the interconnection of computing devices—such as computers, servers, and IoT devices—that enables them to communicate and share data and resources using communication protocols.

1. Key Components of Networking

1.1 IP Address

An IP address is a logical address used to identify a device or network interface within an IP network. It enables devices to identify the source and destination of IP packets during communication. An IP address may be assigned to a host or interface depending on the network configuration.

Two major versions are used:

Version	Description
IPv4	Uses 32 bits, divided into four 8-bit octets. It is normally written in decimal notation, with octets separated by periods. Example: 192.168.1.10
IPv6	Uses 128 bits, divided into eight 16-bit groups. It is written in hexadecimal notation, with groups separated by colons. Example: 2001:db8::1

1.2 Node

A node is a connection point on a network that can send, receive, create, or process data. Depending on the context, a node can include devices such as computers, servers, printers, routers, switches, or IoT devices. A node may be identified using addresses such as an IP address or MAC address.

1.3 Switch

A network switch connects devices within a network and forwards Ethernet frames toward their intended destination. Switches commonly operate at the Data Link layer (Layer 2) of the OSI model, although multilayer switches can also perform Layer 3 functions.

Switching vs. Switch

Switch is a networking device. Switching is the process of forwarding data between network devices or network segments. The term switching can also refer to different communication techniques used to transfer information across networks.

2. Types of Switching

2.1 Circuit Switching

Circuit switching establishes a dedicated communication path between endpoints for the duration of a session. Resources on the path are reserved for that connection. Traditional telephone networks are a common historical example.

2.2 Message Switching

Message switching transfers an entire message from one intermediate node to another. Each node stores the complete message before forwarding it to the next node. This approach can introduce significant delay and storage requirements.

2.3 Packet Switching

Packet switching divides data into smaller units called packets. Packets are transmitted through the network and are reassembled at the destination as required. Modern IP networks primarily use packet switching.

Type	Basic idea
Circuit switching	Dedicated path and reserved resources for a communication session.
Message switching	Complete messages are stored and forwarded between intermediate nodes.
Packet switching	Data is divided into packets that are transmitted across the network.

3. Ports

A port number identifies a logical communication endpoint associated with a service or application on a host. Transport protocols such as TCP and UDP use port numbers to help deliver traffic to the appropriate application or service.

Concept	Example
IP address	Identifies the host or network interface at the IP layer.
Port number	Identifies a logical service or application endpoint on the host.

Examples of commonly encountered ports include TCP 22 for SSH, TCP 80 for HTTP, TCP 443 for HTTPS, and UDP 53 for DNS. Port numbers identify services; they do not by themselves guarantee that a particular service is running on that port.

4. Gateways

A gateway is a device or system that provides a path between networks or acts as an entry and exit point for traffic. In many IP networks, a default gateway is typically a router that forwards traffic destined for other networks. Gateway functions can also involve protocol translation or other forms of interoperability.

5. How Computers Communicate

A layered view of network communication

Computers communicate by following agreed rules called protocols. A protocol defines how data is formatted, addressed, transmitted, received, and interpreted. The TCP/IP protocol suite is widely used for communication across modern networks.

Basic Communication Process

1. An application creates data.
2. Protocols add information needed for delivery and communication.
3. Data is divided into packets when required by the protocol stack.
4. Network devices forward the traffic toward its destination.
5. The receiving system processes the packets and passes the data to the appropriate application.

6. Networking and the OSI Model

The OSI model is a conceptual seven-layer framework used to understand how network communication is organized. It separates communication functions into layers, making network design and troubleshooting easier to understand.

Layer	Name	Examples / Role
7	Application	Network services used by applications; HTTP, DNS, SMTP.
6	Presentation	Data representation, encoding, compression, and encryption concepts.
5	Session	Establishes and manages communication sessions.
4	Transport	End-to-end delivery; TCP and UDP; ports.
3	Network	Logical addressing and routing; IP and routers.
2	Data Link	Frames, MAC addressing, and switching on local networks.
1	Physical	Transmission of bits through cables, radio, fiber, and physical interfaces.

7. TCP/IP Model

The TCP/IP model describes networking using a smaller set of layers and is closely associated with the protocols used on the Internet.

Layer	Examples
Application	HTTP/HTTPS, DNS, SSH, SMTP and other application protocols.
Transport	TCP and UDP.
Internet	IP, ICMP and routing-related functions.

Layer	Examples
Network Access	Ethernet, Wi-Fi and other link/physical technologies.

8. Key Takeaway

Remember

Networking is the foundation that allows systems to communicate. For cybersecurity, understanding IP addresses, nodes, switches, ports, gateways, protocols, packet switching, and layered communication is essential for analyzing traffic, troubleshooting connectivity, understanding attacks, and interpreting security events.

Conclusion

A strong networking foundation helps cybersecurity learners understand what happens when data moves between systems. By learning addressing, switching, routing, ports, protocols, and network layers, you can build the foundation needed for topics such as Nmap, Wireshark, firewalls, IDS/IPS, VPNs, network attacks, and SOC monitoring.
